

1. Administrative & Study Identification

Full Study Title: Evaluating the Efficacy and Safety of PT027 Compared With PT007 Administered As Needed in Participants 12 to < 18 Years of Age With Asthma **Short Title/Acronym:** ACADIA Study **Protocol Number:** D6934C00001 **NCT Number:** NCT06307665 **Sponsor Name:** AstraZeneca **Investigational Product:** PT027 (budesonide/albuterol metered-dose inhaler [BDA MDI]) vs. PT007 (albuterol sulfate metered-dose inhaler [AS MDI]) **Phase of Development:** Phase IIIB **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To compare the effect of budesonide/albuterol metered-dose inhaler (BDA MDI) with albuterol sulfate metered-dose inhaler (AS MDI), both administered as needed, on the annualized rate of severe asthma exacerbations in adolescents. **Secondary Objectives:** To evaluate the time to first severe asthma exacerbation and the annualized total systemic corticosteroid exposure for asthma per participant.

2.2 Background & Rationale To address the need to control asthma exacerbations in adolescents, particularly those with a history of severe exacerbations. Many patients experiencing asthma symptoms use a short-acting beta-2-agonist (SABA), like albuterol, alone as a rescue medicine, which does not address underlying airway inflammation. This leaves adolescents at risk for severe exacerbations, impaired quality of life, and frequent systemic corticosteroid use. This study evaluates combining a bronchodilator (albuterol) and a corticosteroid (budesonide) in a single rescue inhaler to provide faster relief and prevent severe asthma attacks in young people aged 12 to < 18 years.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (PT007 / albuterol sulfate 180 µg) **Blinding:** Double-blind **Randomization:** Randomized, parallel-group

3.2 Planned Enrollment & Duration Target \$N\$: 440 adolescents **Number of Sites:** 109 sites across 4 countries (United States, China, Mexico, and South Africa) **Estimated Study Start:** 20-May-2024 **Estimated Primary Completion:** 13-Oct-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adolescents aged 12 to < 18 years old. 2. Documented clinical diagnosis of asthma with at least one severe asthma exacerbation in the prior 12 months. 3. Receiving scheduled asthma maintenance therapies for at least 3 months with stable dosing for at least the last 1 month (low-to-high-dose inhaled corticosteroid [ICS] or ICS/LABA).

4.2 Exclusion Criteria 1. Life-threatening asthma defined as any history of significant asthma episode(s) requiring intubation, respiratory arrest, or hypoxic seizures. 2. Experienced > 3 severe asthma exacerbations within 12 months before screening. 3. Oral or systemic corticosteroid (OCS/SCS) use within 4 weeks before Visit 1, or chronic use of OCS/SCS ($\geq$ 3 weeks use in the 3 months prior to Visit 1).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: As-needed budesonide/albuterol 180/160 µg or as-needed albuterol 180 µg. **Route of Administration:** Inhalation via pressurized metered-dose inhaler (pMDI). **Duration of Treatment:** 52 weeks.

5.2 Concomitant & Prohibited Therapy Permitted: Continuation of the patient's own maintenance therapy (low- to high-dose ICS-containing maintenance medications with or without other controllers). **Prohibited:** Use of any oral SABAs within one month prior to study entry.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -7	Informed Consent/Assent, Medical History, Pre-screening questionnaires, Peak flow capabilities assessment
Baseline	Day 0	Randomization, First Inhaler Provision, Asthma education
Interim	Weeks 12, 24, 36, 48	Exacerbation Assessment, Concomitant Medication Review, Safety Check
End of Study	Week 52	Final Assessment, Exacerbation records review, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Annualized rate of severe asthma exacerbations. **Measurement Method:** Clinical assessment and documented exacerbation events over the 52-week treatment period.

7.2 Secondary & Exploratory Endpoints 1. Time to first severe asthma exacerbation. 2. Annualized total systemic corticosteroid exposure for asthma per participant.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard reporting of AEs and tracking of asthma exacerbation severity via clinical site follow-ups and patient diaries. **Serious Adverse Events (SAEs):** Standard protocol timeline, typically reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal/External committee managed by AstraZeneca.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for AEs. **Sample Size Justification:** Target $N = 440$. To minimize the number of adolescents exposed to study medications, the treatment effect will be estimated by partially extrapolating results from the patient population in the adult/adolescent MANDALA trial using a Bayesian dynamic borrowing approach. **Handling of Missing Data:** Standard imputation methods (e.g., Multiple Imputation or Bayesian predictive modeling) as defined in the SAP.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite monitoring and remote data checks to ensure proper inhaler technique and trial adherence. **Audit Trail:** Data collected via eCRF locking mechanisms and monitored for timely data clarification.

11. Study Identifiers & Governance

Primary ID: D6934C00001 **Registry Number:** NCT06307665 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** ACADIA **Official Regulatory Title:** Evaluating the Efficacy and Safety of PT027 Compared With PT007 Administered As Needed in Participants 12 to < 18 Years of Age With Asthma

12. Clinical Framework (Asthma Specific)

Primary Condition: Asthma in Adolescents **Diagnosis Threshold:** Documented clinical diagnosis + at least 1 severe exacerbation in the prior year **Medical Methodology:** As-needed combination rescue medication (ICS/SABA) **Optimization Target:** Reduction in severe asthma exacerbation rate and steroid exposure

13. Study Procedures & Methodology

Management Pathway: As-needed rescue inhaler implementation atop stable maintenance therapy **Education Protocol (HCP):** Site initiation visits and investigational product administration training **Education Protocol (Patient):** Inhaler technique instruction (without spacers) and reliable peak flow measurement training **Quality Audit Mechanism:** Assessment of appropriate inhaler use and continuous maintenance therapy tracking

14. Observation & Follow-Up Schedule

Total Duration: 52 weeks (plus up to 28 days of screening) **Onsite Visit Cadence:** Scheduled clinic visits throughout the 52-week treatment period **Remote Monitoring Frequency:** Intermittent study site calls and check-ins **Checklist Review Interval:** Regular recording of asthma symptoms and exacerbation events

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				